higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(First Paper**)**
NQF LEVEL 2

(10501042)

21 February 2019 (X-Paper)
09:00–12:00

Non-programmable scientific calculators may be used.

This question paper consists of 11 pages, 1 formulae sheet and 1 answer sheet.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Questions may be answered in any sequence. Subsections of questions may NOT be separated.
5. Show ALL the calculations and the intermediary steps.
6. Simplify answers where possible.
7. ALL final answers must be approximated accurately to THREE decimal places.
8. The list on the formula sheet is NOT necessarily complete. Any other applicable formulae may be used.
9. Diagrams are NOT drawn to scale.
10. Write neatly and legibly.

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QUESTION 1

1.1 Various possible options are provided as answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Given: $x = \dfrac{16-[-8\times(15-13)]}{4}$. The value of x is:

A 8
B 16
C 20
D 64

1.1.2 The best classification for -6 is:

A Natural number, real number and an integer
B Integer, rational number and a real number
C Irrational number and a real number
D Integer, irrational number and a real number

1.1.3 $\sqrt[3]{8x^5}$ expressed in exponential form is:

A $8x^{\frac{3}{5}}$
B $\dfrac{8}{3}x^{\frac{5}{3}}$
C $2x^{\frac{5}{3}}$
D $2x^{\frac{3}{5}}$

1.1.4 The first term of an arithmetic sequence is 4 and the common difference is 7. The first 4 terms of this arithmetic sequence are:

A 4; 11; 18; 25
B 7; 11; 15; 19
C 4; 8; 12; 16
D 4; 11; 14; 16

1.1.5 The number line below represents the following inequality:

A $\{x|x\in R;-2\le x<3\}$
B $\{x|x\in R;-3\le x<2\}$
C $\{x|x\in R;-2\ge x>3\}$
D $\{x|x\in R;-3<x\le 2\}$ (5)

1.2 Convert the decimal fractions below to the form $\frac{a}{b}$; where a; b$\in$Z and b$\neq$0.

Express your answer in its simplest form.

1.2.1 1,25 (1)

1.2.2 $2,9\dot{4}\dot{3}$ (3)

1.3 Simplify the following by using the surd laws.
Show the expansion of surds and all intermediate steps.

$$\frac{7\sqrt{108}-4\sqrt{75}-3\sqrt{48}}{-\sqrt{27}}$$ (4)

1.4 Simplify the following by using the laws of exponents.
Leave answers with *positive* exponents and in *surd* form where applicable.

1.4.1 $5(x+y)^0$ (1)

1.4.2 $\frac{2x^2y\times 4xy^{-3}}{8xy^{-2}}$ (2)

1.4.3 $\left(\frac{(49a^{-6})^{\frac{1}{2}}}{\sqrt{81b^{-6}}}\right)^{-1}$ (3)

1.5 Given: $A_t = A_0\left(1+\frac{r}{100}\right)^n$

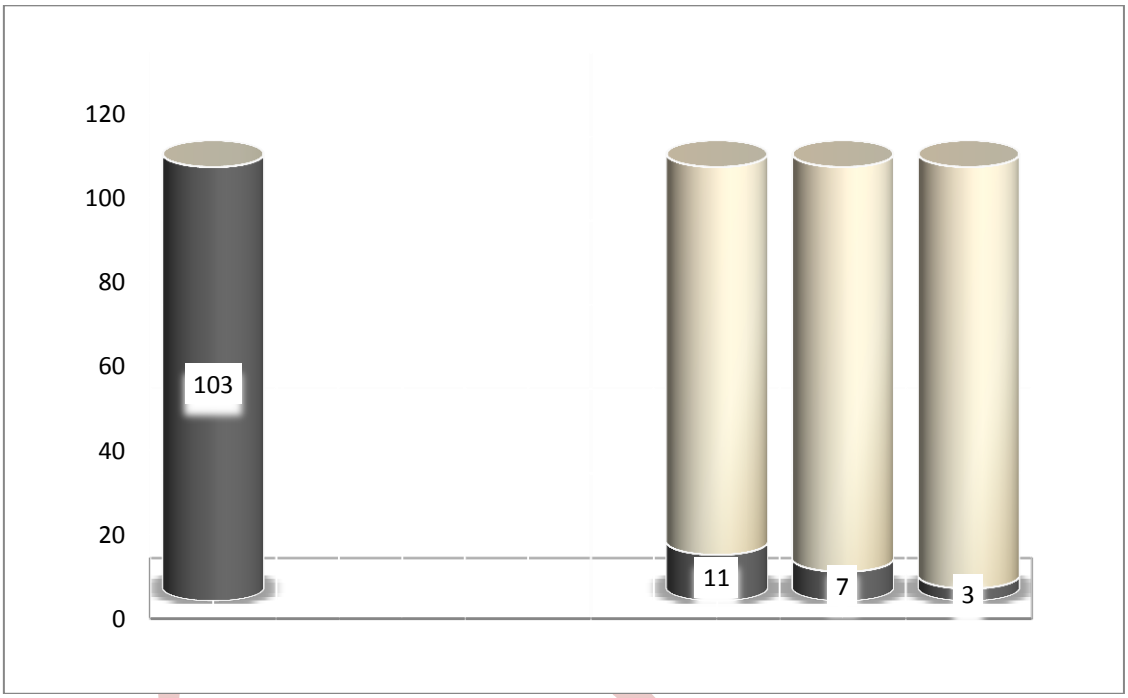
1.5.1 Make 'r' the subject of the formula.

1.5.2 If $n=4$, $A_0=3\,000$ and $A_t=4\,200$.
Calculate the value of r correct to two decimals.

(2 × 2) (4)

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1.6 The volume of water in a cylindric reservoir tank represents an arithmetic sequence. On the first day there is 103 kilolitres. The volume decreases over time until only 3 kiloliters are left in the tank.The diagram below shows the volume of water on the first day (103) and any three consecutive days after (11; 7; 3).



1.6.1 Determine the level of water on the fifth day (T_5). (2)

1.6.2 Determine on which day (the value for n) the water level will be at 3 kilolitres. (3)

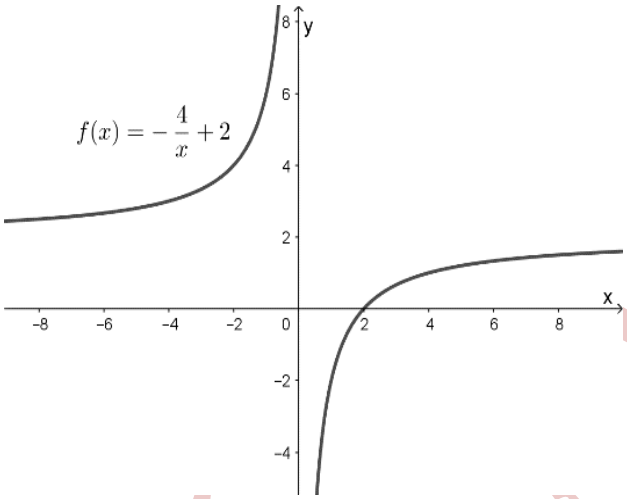
1.6.3 Determine if there will be water left on the 28th day if the usage of water stays constant. Show ALL calculations. (2)

[30]

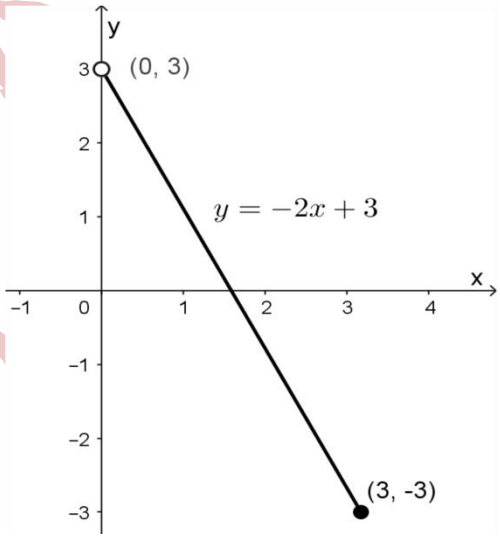
QUESTION 2

2.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (2.1.1–2.1.6) in the ANSWER BOOK.

Study the graphs given below and answer the questions.



GRAPH 1



GRAPH 2

2.1.1 Which statement correctly describes the horizontal asymptote for Graph 1:

A $y = 2$
B $y = 0$
C $x = 0$
D $x = 2$

2.1.2 Which of the following describes the domain for Graph 1 the best:

A $y \in \mathbb{R}$
B $x \in [-\infty; 0)$ *and* $(0; \infty]$
C $x \in (-\infty; 0)$ *and* $(0; \infty)$
D $x \in \mathbb{R}$

2.1.3 Which of the following statements is true for Graph 1:

A $f(x)$ is continuous
B $f(x)$ is discontinuous
C $f(x)$ has a y- intercept
D The name of $f(x)$ is a parabola

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2.1.4 In the graph $y = \frac{a}{x} + q$ an increase in the value of q will cause a:

A horizontal transformation in the graph to the left
B horizontal transformation in the graph to the right
C vertical transformation in the graph upwards
D vertical transformation in the graph downwards

2.1.5 Which domain and range below corresponds with the given constraints on Graph 2:

A Domain: $\{x|0 < x \leq 3; x \in \square\}$ and range: $\{y|-3 < y \leq 3; y \in \square\}$
B Domain: $\{x|0 < x \leq 3; x \in \square\}$ and range: $\{y|-3 \leq y < 3; y \in \square\}$
C Domain: $\{x|x > 0; x \in \square\}$ and range: $\{y|-3 < y < 3; y \in \square\}$
D Domain: $\{x|x > 0; x \in \square\}$ and range: $\{y|-3 \leq y < 3; y \in \square\}$

2.1.6 The magnitude of the gradient for Graph 2 is:

A $m = \frac{x_2 - x_1}{y_2 - y_1}$
B $m = 3$
C $m = -2x$
D $m = -2$

(6 × 1) (6)

2.2 Given: $f(x) = 2x^2 - 4$

2.2.1 Answer this question on the attached ANSWER SHEET.

Sketch the graph $f(x) = 2x^2 - 4$ for the domain $[x|x \in R; -2 \leq x < 2]$ by using the table method. Clearly show coordinates and the endpoints of the graph. (4)

2.2.2 What will the effect on the graph be if the sign of the coefficient of x^2 changes from 2 to -2? (1)

2.3 Answer this question (2.3.1 AND 2.3.2) on the attached ANSWER SHEET.

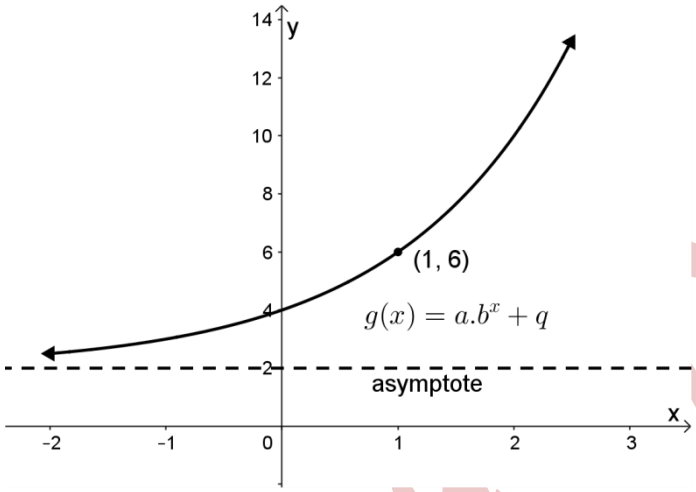
Given: $y = -x + 5$ and $y = x - 3$.

2.3.1 Sketch both graphs on the same system of axis. Clearly show the coordinates of the intercepts with the axis. (4)

2.3.2 Hence, determine the simultaneous solution for the two graphs graphically and indicate it on the graph. (2)

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2.4 The graph below passes through the points (1; 6) and (0; 4).



2.4.1 Determine the values for a, b and q and give the equation for $g(x)$. (3)

2.4.2 What is the name of the graph $g(x)$? (1)

2.4.3 Give the range for $g(x)$. (2)

2.4.4 Is $g(x)$ a function or a non-function? (1)

2.4.5 Give a reason for your answer in QUESTION 2.4.4. (1)

[25]

QUESTION 3

3.1 Simplify the following:

3.1.1 $(2x-3)(x^2-x+1)$

3.1.2 $2x-4x(2x-5)-(3x+21)$

(2 × 2) (4)

3.2 Factorise the following expressions:

3.2.1 a^3-ab^2 (2)

3.2.2 $12x^2-20x+7$ (2)

3.2.3 $25p^2-16+(5p+4)(p-2)$ (3)

3.3 Simplify the following:

$$\frac{7xy\ -21x^2y^2+21x^3y^3}{7xy}$$ (2)

3.4 Solve for x in the following equations:

3.4.1 $\frac{1}{4}=\frac{x-3}{8}$

3.4.2 $(x-2)(x+2)=3x^2+x-10$

3.4.3 $9^{2x+3}=27^{x+5}$

(3 × 2) (6)

3.5 Given: $-2(x+3)>5x-76$

3.5.1 Solve the inequality. (2)

3.5.2 Represent the solution in interval notation. (1)

3.6 Solve the following system of linear equations algebraically.

$$3x+4y=4$$
$$6x-y=26$$

(3)
[25]

QUESTION 4

4.1 Give ONE word or term for each of the following descriptions. Write only the word/term next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

4.1.1 A card issued by a bank or business that allows somebody to purchase goods and services and pay for them later, often with interest

4.1.2 Expenses that stays the same and are paid regularly

4.1.3 A money lender who lends out his/her own money

4.1.4 It is a plan, fund or program which provides retirement income to employees

4.1.5 The difference between the actual and projected amounts

(5 × 1) (5)

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4.2 The following table shows Janes actual budget.

Study the table and answer the questions.

Quarterly Budget					
	Jan-March	**April-June**	**July-September**	**October-December**	**TOTAL**
INCOME					
Salary	R4 900,00	R4 900,00	R4 900,00	R4 900,00	R19 600,00
Interest on investment	R75,00	R152,25	R213,82	R313,77	R754,84
TOTAL	**R4 975,00**	**R5 052,25**	**R 5113,82**	**R5 213,77**	**R20 354,84**
EXPENSES					
Rent	R2 850,00	R2 850,00	R2 850,00	R2 850,00	R11 400,00
Taxi	R580,00	R370,00	R470,00	R370,00	R1 790,00
Food	R1 450,00	R520,00	R1 258,82	R1 130,00	R4 358,82
Entertainment	R30,00	R1 192,25	R215,00	R200,00	R1 637,25
Cell phone	R320,00	R320,00	R320,00	R320,00	R1 280,00
TOTAL	R5 230,00	R5 252,25	R5 113,82	R4 870,00	**R20 466,07**
Variance	**-R255,00**	**-R200,00**	**R00,00**	**R343,77**	**A**

4.2.1 Calculate the closing balance (A) at the end of the period.

4.2.2 During which period was there a surplus?

4.2.3 Name the income that is a fixed income.

4.2.4 What type of interest was used on the investment?

(4 × 1) (4)

4.3 R20 000 is invested in a JSE stock exchange for eight years. The growth of the investment is on average 12% p.a. during that period.

Calculate the accrued amount under the following conditions:

4.3.1 Simple interest

4.3.2 Compound interest calculated annually

(2 × 2) (4)

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4.4 White bread at the local supermarket costs R12,99 per loaf.

4.4.1 Your grandmother told you that bread was very cheap 50 years ago. If the inflation rate was on average 9,17% over the period, show that the price of bread 50 years ago was 16 cents per loaf. (3)

4.4.2 The price of bread 50 years ago was 16 c and your grandmother's salary was R2,00 per day. Your current daily salary is R160,00 per day and the current bread price is R12,99.

Calculate the percentage of the daily salary spent on bread, both for your grandmother and yourself. (3)

4.4.3 What can you deduce from the answer in QUESTION 4.4.2. (1)
[20]

TOTAL: 100

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FORMULA SHEET

1) $a^m \times a^n = a^{m+n}$

2) $a^m \div a^n = a^{m-n}$

3) $\left(a^m\right)^n = a^{m \times n}$

4) $\left(a^m b^n\right)^p = a^{mp} . b^{np}$

5) $\left(\frac{a^m}{b^n}\right)^p = \frac{a^{mp}}{b^{np}}$

6) $a^{-n} = \frac{1}{a^n}$

7) $a^0 = 1$

8) $\sqrt[n]{a^m} = a^{\frac{m}{n}}$

9) $T_n = a + (n-1)d$

10) $S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

11) $S_n = \frac{n}{2}(a + l)$

12) $I = A_0 \times \frac{r}{100} \times t \quad or \quad I = \frac{\Pr t}{100} \quad or \quad A_1 = P(1+i)^n$

13) $A_t = A_0\left(1 + \frac{r}{100 \times m}\right)^{t \times m} \quad or \quad A_t = P(1+i)^n$

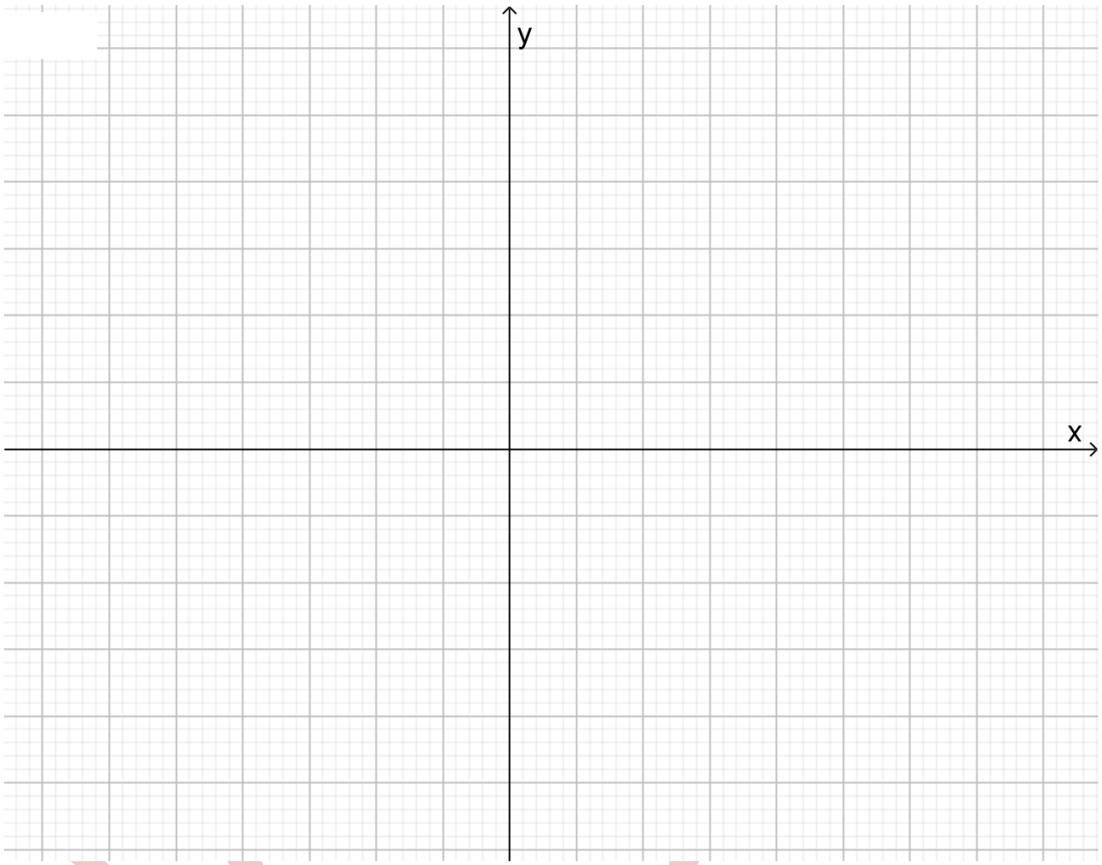
14) $i = \frac{r}{100}$

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ANSWER SHEET EXAMINATION NUMBER:

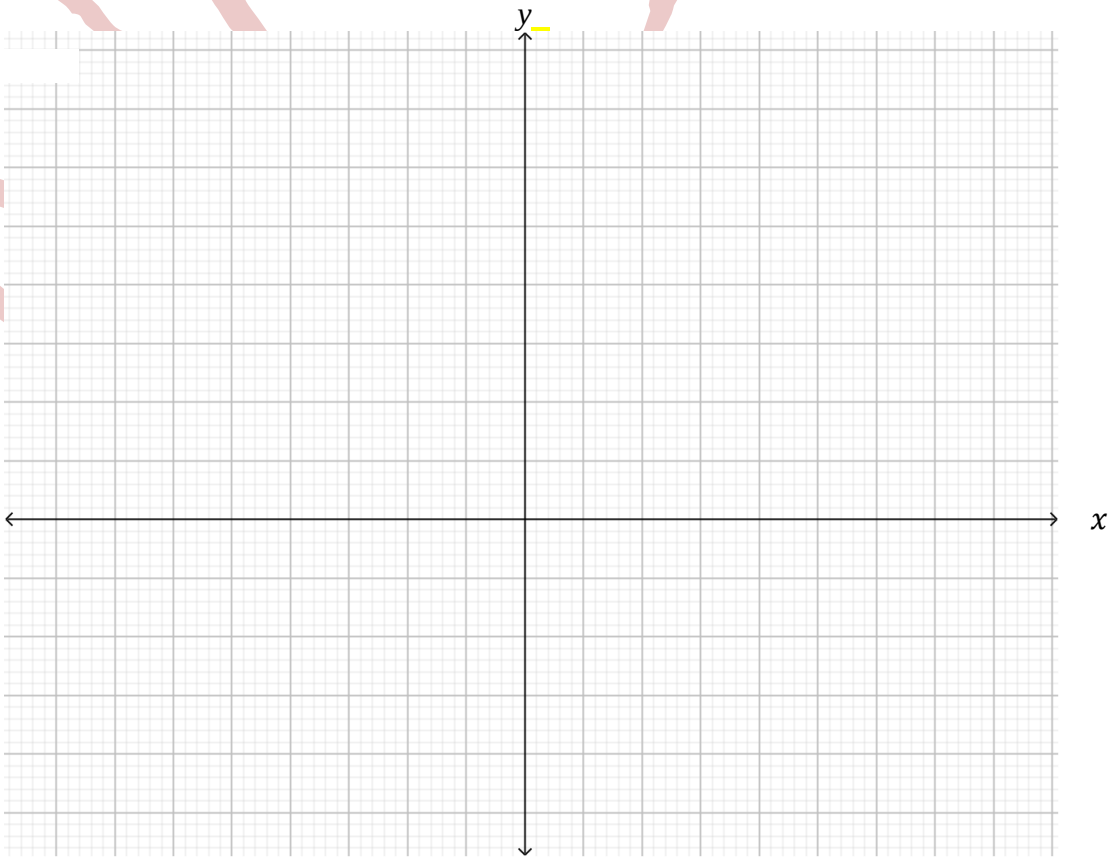
2.2

2.2.1



(4)

2.3.1



(6)

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